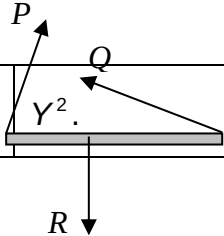


28	The de Broglie wavelength of a particle that has kinetic energy Y is λ. Given that the mass of the particle remains constant, the wavelength λ is directly proportional to			
	A $\frac{1}{Y^2}$	B $\frac{1}{Y}$	C $\frac{1}{\sqrt{Y}}$	 A diagram showing a horizontal grey bar representing a surface. At the left end of the bar, a vector P points upwards and to the right. At a point Q on the bar, a vector points upwards and to the left. Below the bar, a vector R points straight down. The label Y^2 is placed near the left end of the bar.

Answer: C

de Broglie wavelength:

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

$$Y = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

$$p = \sqrt{2mY}$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mY}}$$

Therefore, wavelength is directly proportional to $\frac{1}{\sqrt{Y}}$.